

Jonathan Liu

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https://github.com/spooketti
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Education

Purdue University	West Lafayette
B.S. Robotic Engineering Technology	2026-2030
Del Norte High School	San Diego
High School Diploma	2022-2026
Certification in Software and Systems Development	
Certification in Engineering Design	

Skills

Hard Skills

- REST API (JavaScript, Python Flask, Go)
- Data Analysis (OpenCV, CUDA, MatLab, Linux)
- Robotics (Java, Arduino C++, Autodesk Eagle/CAD)
- Game Dev (C#, Unity/Godot, Lua, Blender, Photoshop)

Soft Skills

- Agile teamwork
- Code Review (Pull Requests, Git, Documentation)
- Presenting Results (Abstracts, Conferences, Presentations)

Awards

Del Norte High School	CollegeBoard
• AP Scholar with Distinction Award	
• National Recognition Program - School Recognition Award for Outstanding Academic Achievement	
Research Conference	UCSD OPALS
• ICBEB 2025 Best Poster Award	
Runtime Terror 22105	FTC Robotics
• Inspire 3 Award at 2025 Michiana Premier Event	
• Thompson division finalist alliance (Michiana)	
• 1st place alliance in 2026 San Diego Gauss League Championships	
• Inspire 1 at 2026 San Diego League Championships	
• Inspire 2 at 2026 San Diego Regional, advanced to World Championships	
Team Optix 3749	FRC Robotics
• On The Fly won 2025 Innovation In Control Award at San Diego Regional	

Work Experience

FTC Robot Programming	2025-Present
Runtime Terror 22105	On-Site
• Implemented camera AprilTag relocalization / color masking pipelines	
• Programming endgame hanging in 2025/2026, ball sorting, etc.	
• 2026 World Championships, 11th Qualifier, Edison Playoffs	
Phylogenetic Research Intern	2025 Summer
UCLA Pellegrini Lab	Remote
• Analyzed Hypocreales taxonomy / phylogenetic groups	
• Compare UCLA UMap vs NCBI database .newick trees	
Surgery System Research Intern	2024-2025
UCSD OPALS	Hybrid
• Programming AR spinal tracking surgery navigation system	
• C# ArUco computer vision for localization	
CompSci Education Research Intern	2024-2025
UCSD Mentor Assistance Program	Hybrid
• Analyze K-12 Computer Science Education trends/data	
• Data obtained via Python web-scraping bot	
Library Volunteer	2024-Present
4S Ranch San Diego County Library	On-Site
• Shelving books, customer service, supporting school clubs	
• Achieved 23 volunteer hours/200+ people supported	
FRC Robot Programming	2022-2025
Team Optix 3749	On-Site
• Teaching incoming members Java	
• Repairing software at competitions	
▸ San Diego Regional 2024/2025	
▸ Battle At The Border 2023/2024 competitions	
• Dynamic "On The Fly" path generation with autonomous obstacle avoidance	
• Applied control theory (PID/FeedForward) for motor function	
• 2025 Postseason Differential Drive GoKart	
Personal Projects	
Indie Dev Group	Superposition Development
• Spectator: WIP procedurally generated horror game using Godot with C#	
• Relay: Open source full stack chat system emphasizing privacy via self-hosted web servers	
▸ Webserver using Go/SQL, other versions in Python/C/etc.	
▸ Web client with plans of a Go/C CLI client	
Personal Projects	spooketti on GitHub
• Wearable Arduino Christmas Tree Hat via Breadboard, LEDs, C++, portable battery	
• YT-RPC: YT Music client enabling listen together via WebRTC and status broadcasts to Discord	
Publications/Conferences	
UCSD OPALS	
• Abstract: Development of an Augmented Reality Surgical Navigation System Based on Multiple ArUco Markers	
• Poster (Paper ID: BEB8008) presented at 2025 Japan BEB Conference on Biomedical Engineering and Biotechnology	
UCSD MAP	
• Abstract: Insight On K-12 Computer Science Education Trends Through Scratch Project Analysis	
• Presented at 2025 USD Consortium For Computing Sciences In Colleges Conference	